Daniel Yoo

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Education

University of Toronto **Expected May 2028**
B.A.Sc. in Mechanical Engineering (Mechatronics and Energy) *Toronto, ON*
Relevant Coursework: Probability & Statistics for Engineering Analysis, Mechanical Design, Manufacturing Engineering, Numerical Analysis & Computational Methods, Analog and Digital Electronics for Mechatronics, Engineering Economics, Urban Engineering Ecology, Fundamentals of Computer Programming, Solid Mechanics

Technical Skills

Data & Analysis: SQL (data querying), Tableau (data visualization), Microsoft Excel (data organization, tracking, automation), MATLAB, Data validation, structured data processing, engineering documentation, Visio
Programming & Tools: Python (data processing, automation), Microsoft Office (Excel, Word, PowerPoint)
Engineering & Design: SolidWorks (CSWA Certified), ANSYS Workbench, OpenLCA, EIO-LCA
Electronics & Simulation: LTspice (circuit simulation), Autodesk Eagle (PCB design)
License: Ontario G Driver License

Experience

Engineering Workshop Lead **May 2025 – Present**
Engineers Without Borders: UofT Chapter *Toronto, ON*

* Led 6+ engineering workshops for over 60 students, guiding **structured problem-solving and system-level design**
* Developed and delivered **technical presentations and structured worksheets** to improve engagement and understanding
* Managed **budget tracking and materials** using Excel, supporting multiple sessions with consistent cost control
* Coordinated **team activities and facilitated collaboration** in group-based engineering challenges

Fire Direction Center (FDC) Specialist **Nov 2023 – May 2025**
Republic of Korea Army — Artillery Unit *Gyeonggi, South Korea*

* Developed **Excel and VBA-based automation tools** to streamline artillery calculations, reducing manual processing time by approximately 40%
* Managed and validated **100+ data entries per operation** to ensure accuracy in **time-critical decision-making**
* Improved calculation accuracy and reduced human error through **structured data verification processes**
* Integrated and processed multiple data inputs to support coordinated system-level operations

Technical Projects

Electricity Demand Analysis & Visualization (Python, SQL, Tableau) | GitHub **Apr 2026 – Apr 2026**

* Processed and analyzed **large-scale time-series data** (**approximately 100,000+ hourly records**) using **Python and SQL** to extract temporal demand patterns
* Built **interactive Tableau dashboards** to visualize demand trends across hourly, monthly, and weekday dimensions
* Identified **peak demand periods reaching approximately 53,000 MW**, with clear daily cycles and evening peak usage trends
* Applied **time-series feature engineering** to quantify demand variation across time, enhancing pattern detection and enabling more accurate demand analysis

Solar-Assisted LED Desk Lamp System Design (OpenLCA, EIO-LCA) | GitHub **Jan 2026 – Apr 2026**

* Redesigned an LED desk lamp system integrating **solar panels and battery storage** to improve energy efficiency.
* Collected and analyzed energy performance data, reducing grid demand from **9.13 to 6.81 kWh/year** (25% reduction).
* Prepared technical and financial evaluations, including Life Cycle Assessment (LCA), payback period, and NPV to support design decisions.
* Evaluated **engineering trade-offs** between environmental impact, cost, and system performance, presenting data-driven recommendations.

PCB Pick-and-Place Machine (SolidWorks) | GitHub **Sep 2022 – Dec 2022**

* Designed a **600-mm desktop PCB Pick-and-Place Machine** integrating a rack-and-pinion Z-axis and belt-driven X–Y gantry system for precise component positioning.
* Built a complete SolidWorks mechanical assembly (100+ components) and generated manufacturing-ready **engineering drawings** and **assembly documentation.**
* Developed and validated **mechanical test** setups to evaluate positioning accuracy, repeatability, and subsystem integration performance.
* Developed a **BOM** covering 60+ components across 5 subsystems and performed tolerance analysis to improve alignment and assembly reliability.